

Red vs. Blue: A Cyber Defense Game That Teaches Itself

Design brief for the UI team. No tech background needed.

The idea

Think of the chess computer that got world-class by playing against itself millions of times. We do the same thing for cybersecurity. An AI **attacker (Red)** tries to break into a pretend computer network. An AI **defender (Blue)** tries to stop it. They play over and over, and each game Blue gets a little better at protecting the network. Your job: make that battle easy and fun to watch.

The board

The network is a map of dots (**nodes**) connected by lines. Each node is a device: laptops, a router, a printer, a server, and one **crown jewel** (the database with the valuable stuff). Lines show which devices can talk to each other. Red can only move along lines, one hop at a time.

How a turn works

Step	What happens
1. Red moves	Red picks one attack against one node.
2. Blue moves	Blue picks one defense. It may or may not have noticed Red yet.
3. Map updates	Node colors change to show who controls what.
4. Log updates	Both moves appear in the log in plain English.
5. Repeat	About 20 turns per game. Then a new game starts, and both sides are a bit smarter.

How someone wins

Red wins by taking over nodes: it sneaks in at the edge (usually a laptop), then hops from node to node along the lines, grabbing passwords and access as it goes. Red wins if it **captures the crown jewel** or **takes over half the network**.

Blue wins if it **survives all 20 turns** without losing the crown jewel, or **kicks Red out of every node**. Blue has one catch: it can't always see Red. It has to spend turns looking.

Example attacks (Red)

Attack	Plain English
Phishing	Tricks a person into clicking a bad link. Usual way in.
Password guessing	Tries common passwords until one works.
Exploit old software	Uses a known flaw in a device that hasn't been updated.

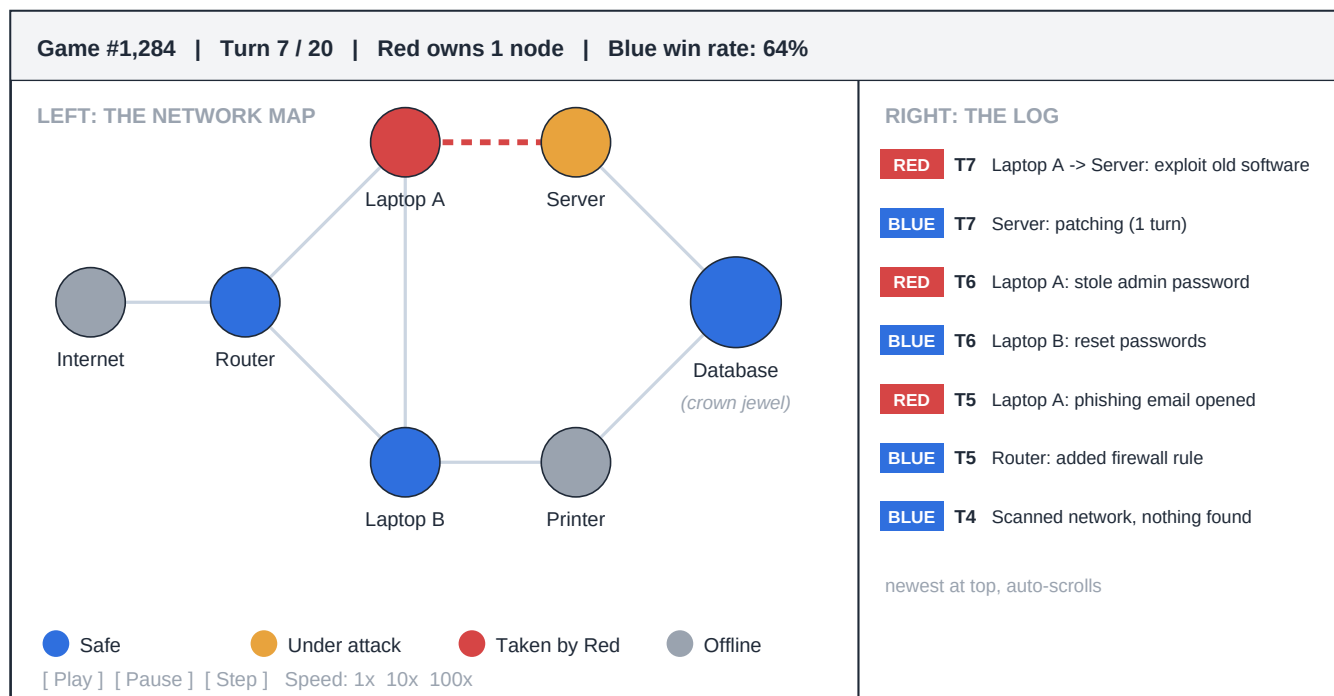
Steal passwords	Once inside, grabs saved logins to unlock more nodes.
Move sideways	Hops to a connected node using what it's stolen.
Steal data	Copies files off a node. Game over if it's the crown jewel.

Example defenses (Blue)

Defense	Plain English
Scan	Looks for signs of Red. Reveals hidden attacks.
Patch	Updates software so old flaws stop working. Takes a turn.
Reset passwords	Makes stolen passwords useless.
Firewall rule	Blocks a line so Red can't travel that way.
Isolate node	Unplugs a device. Stops spread, but it's offline too.
Restore from backup	Wipes a taken node and gives it back to Blue.

Screen layout

Left (about two thirds): the live network map. **Right (about one third):** the log, a running play-by-play. **Top bar:** game number, turn, score. **Bottom:** play, pause, step, and speed.



Sample log

```
Game #1,284
T1 RED Sent phishing email to Laptop A
T1 BLUE Scanned Router: all clear
T2 RED Laptop A clicked the link. Red is in.
T2 BLUE Patched Server
T3 RED Stole admin password on Laptop A
T3 BLUE Scanned Laptop A: intruder found!
T4 RED Tried Laptop A -> Server: blocked (already patched)
T4 BLUE Isolated Laptop A. Red is cut off.
RESULT Blue wins. Red held 1 node for 2 turns.
```

Design wishlist

- **Color = ownership.** Blue safe, amber under attack, red taken, gray offline. Readable at a glance.
- **Show movement.** A red pulse traveling down a line when Red hops. A shield flash when Blue blocks.
- **Hidden vs. seen.** Red's sneaky moves could appear faintly until Blue scans and spots them.
- **Log and map linked.** Hover a log line and its node lights up (and the reverse).
- **Show learning.** A small chart of Blue's win rate climbing over thousands of games. This is the "wow" moment.
- **Speed control.** 1x to follow a single game, 100x to watch the AI improve.